Practice 1

1 a Draw the next two shapes in the sequence below.



b Explain the pattern in words.

c Write this pattern as a sequence of five numbers.
How do you find the next number in the sequence?

d Is this a linear pattern? Explain.

e If one of the shapes has L leaves in it, how many leaves will there be in the next shape?

f Determine what the 10th term in the sequence would be, without drawing it.
How do you know?

g Will the number 30 be in this sequence? Explain why or why not.

2 a Draw the next two shapes in the sequence below.



b Explain the pattern in words. How do you find the next number in the sequence?

c Is this a linear pattern? Explain.

d Determine what the 30th term in the sequence would be, without drawing it. How do you know?

e If one of the shapes has w water droplets in it, how many droplets will there be in the next shape?

f Will the number 40 be in this sequence? Explain why or why not.

Definition

Algebraic Expression — An expression consisting of numbers, constants, variables, and operation symbols.

Examples of variable expressions:

$$3x$$

$$4x^2 + 1$$

$$-3x^3y + y^2z$$

$$2\pi r$$